

JASON FORCE

Spacecraft Systems • Optical Payloads • Firmware & Controls

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Professional Summary

Space systems engineer with experience spanning spacecraft subsystem design, optical payload technologies, firmware, controls, sensing, and precision electromechanical systems. Experienced developing and integrating spacecraft-related software, optics and laser systems, motion systems, and multidisciplinary sensing platforms for aerospace and advanced technical applications. Background includes spacecraft electrical power systems, lidar and optical tracking technologies, precision pointing mechanisms, and real-world system integration.

Core Technical Expertise

Spacecraft Systems: Spacecraft subsystem design, bus-to-payload interfaces, system architecture trade studies, subsystem coordination, spacecraft software

Optical Payloads & Sensing: Lidar systems, laser systems, optics, passive optical tracking, computer vision target tracking, motion estimation, sensing systems

Controls & Estimation: Kalman filtering, factor graphs, state estimation, sensor fusion, feedback control, precision motion systems

Software & Firmware: Python, C++, firmware, Embedded Linux, RTOS, ARM architectures, asynchronous systems, real-time control

Optics, Motion & Mechatronics: Optomechanical design, precision motion systems, frameless motors, motor drive systems, precision bearings, pointing systems, electromechanical systems

Robotics & Navigation: ROS 2, VINS-Fusion, PX4, MAVROS, MAVLink, Gazebo, GTSAM with ISAM, SLAM, autonomous navigation

Hardware & Processing: CPU/GPU/TPU/DSP/FPGA architectures, PCB-based electronics, microprocessors, AI coprocessors, Intel RealSense, Ambarella, Hailo

Engineering & Development Tools: Git, VSCode, PyCharm, Jupyter Notebooks, Google Colab, Amazon SageMaker, Claude, Siemens NX

Technical Communications & Planning: Technical documentation, system architecture communication, Confluence, Jira, EaseRequirements, requirements traceability, multidisciplinary technical coordination

Professional Experience

Software Lead

Overview Energy | Sep 2024 – Present

- Technical lead in an early-stage space systems startup supporting spacecraft software, subsystem definition, and multidisciplinary technical development for space-based energy systems.
- Participate in spacecraft communications and bus-to-payload interface trade studies.
- Support optical payload and precision motion system development involving high-power laser systems, optics, frameless motors, precision bearings, and pointing-related mechanisms.
- Develop firmware and software interfaces for spacecraft-related subsystems using CAN, SPI, I2C, and SSI encoder interfaces.
- Perform optomechanical and mechanical design activities using Siemens NX.

Mobile Robotics Perception Engineer

Force Consulting Services LLC | Dec 2023 – Sept 2024

- Developed and evaluated visual inertial navigation system (VINS) approaches for UAV localization and navigation in varied operating environments.
- Developed geo-referenced imagery navigation and VINS-based SLAM systems.
- Integrated ROS 2, PX4, MAVROS, MAVLink, and Gazebo-based simulation and autonomy workflows.
- Evaluated perception, target tracking, and autonomy approaches for robotics and navigation applications.

Machine Vision Engineer

Force Consulting Services LLC for United States Postal Service | Nov 2020 – May 2023

- Developed AI and computer vision algorithms for logistics and automation applications.
- Evaluated ORB-SLAM3 for facility navigation and robotic guidance.
- Developed machine vision systems for perception, tracking, and sensing applications.
- Built perception pipelines using Python, OpenCV, Open3D, asyncio, and related tooling.

Director of Imaging Systems

Applied Signals Intelligence Inc. | May 2019 – Nov 2020

- Developed UAV vision-based pose estimation software using ground fiducials.
- Created software interfaces for camera systems and optical tracking hardware.
- Developed image processing applications on Raspberry Pi platforms.
- Supported passive optical tracking and navigation applications.

President & Principal Engineer

Force Labs Inc. | Feb 2012 – Aug 2019

- Led multidisciplinary engineering efforts involving optics, laser systems, firmware, sensing, controls, and precision electromechanical systems.
- Developed lidar, laser, and precision pointing subsystems for government and aerospace-related applications, including fast steering mirrors, polygon scanning systems, and feedback control architectures.
- Architected firmware, PCB-based electronics, software systems, sensing, and electromechanical platforms for advanced technical applications.
- Managed technical proposals, engineering coordination, and system-level architecture decisions.

Electrical Integration Engineer

Orbital Sciences Corporation | 1996 – 1997

- Supported electrical power subsystem development for communications satellite systems.
- Assisted hardware and software design changes for spacecraft electrical power systems.
- Served as shift test director during thermal vacuum testing and managed engineering development unit activities.
- Developed verification procedures and electrical power subsystem software specifications.

Education

M.S., Electrical Engineering — George Mason University, Fairfax, VA (2017)

B.S., Electrical Engineering — George Mason University, Fairfax, VA (1995)

Professional Development

Selected coursework in computer vision, AI, robotics, controls, and related technologies.